Chapter 7 — Reaction Kinetics

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 158–162

Q.1 — Multiple Choice Questions (with correct option and reason)

I. The rate of reaction:

(a) Increases as the reaction proceeds
(b) Decreases as the reaction proceeds
(c) Remains the same as the reaction proceeds
(d) May decrease or increase as the reaction proceeds

Answer: (b) Decreases as the reaction proceeds — As reactants are consumed their concentrations fall, so the rate generally decreases with time.

II. Increasing the temperature increases the rate because:

(a) Both the collision frequency and collision energies of reactant molecules increase
(b) Collision frequency of reactant molecules increases
(c) Activation energy increase
(d) Activation energy decrease

Answer: (a) Both the collision frequency and collision energies of reactant molecules increase — Higher temperature increases both how often molecules collide and the fraction with energy ≥ Ea.

III. Two reactions with different activation energies at the same temperature: the one with lower Ea has:

(a) A smaller rate constant
(b) A larger rate constant
(c) The same rate constant
(d) A rate constant that depends on the enthalpy change

Answer: (b) A larger rate constant — From k = Ae^(−Ea/RT), a lower Ea gives a larger rate constant.

IV. The order of a chemical reaction that is independent of concentration is:

(a) Second order reaction
(b) First order reaction
(c) Zero order reaction
(d) Pseudo first order reaction

Answer: (c) Zero order reaction — For a zero-order reaction rate = k, independent of reactant concentration.

V. On a Boltzmann distribution curve, the area under the curve represents:

(a) Activation energy of the reaction
(b) Total number of molecules in the sample
(c) Average kinetic energy of the molecules
(d) Rate constant of the reaction

Answer: (b) Total number of molecules in the sample — The total area corresponds to the total number of molecules.

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VI. On a Boltzmann distribution curve, the activation energy (Ea) is represented by:

(a) The height of the peak
(b) The area under the entire curve
(c) A vertical line drawn at a specific kinetic energy value
(d) The difference between the peak and the X-axis

Answer: (c) A vertical line drawn at a specific kinetic energy value — Ea is marked as a vertical line at the minimum energy required for reaction.

VII. If doubling the concentration of a reactant increases the rate four times, the reaction is:

(a) Second order
(b) First order
(c) Third order
(d) Zero order

Answer: (a) Second order — Rate $\propto [A]^2$ because $2^2 = 4$.

VIII. The rate determining step in a multi-step reaction is:

(a) Always the first step
(b) Always the last step
(c) The slowest step
(d) The fastest step

Answer: (c) The slowest step — The slowest step limits the overall rate.

IX. For $NO_2 + CO \rightarrow NO + CO_2$ via slow $2NO_2 \rightarrow N_2 + NO_3$ (k_1) and fast $NO_3 + CO \rightarrow CO_2 + NO_2$ (k_2), the rate law is:

(a) $R = k_1[NO_2]^3$
(b) $R = k_2[NO_3][CO]$
(c) $R = k_1[NO_2]$
(d) $R = k_1[NO_2]^2$

Answer: (d) $R = k_1[NO_2]^2$ — The slow step is bimolecular in NO_2, so rate = $k_1[NO_2]^2$.

X. How does a catalyst affect the rate of a chemical reaction?

(a) It always decreases the rate.
(b) It always increases the rate.
(c) It increases the forward rate and decreases the reverse rate.
(d) It increases the rate of both forward and reverse reactions.

Answer: (d) It increases the rate of both forward and reverse reactions. — A catalyst lowers Ea for both directions, speeding up forward and reverse reactions equally.

XI. On an energy profile diagram, a catalyst is represented by:

(a) A higher peak representing the activation energy
(b) A lower peak representing the activation energy
(c) A change in the energy level of reactants or products
(d) A shift in the equilibrium position

Answer: (b) A lower peak representing the activation energy — A catalyst provides an alternative pathway with lower activation energy (lower peak).

XII. The units of the rate constant k depend on the:

(a) Activation energy of the reaction
(b) Temperature of the reaction
(c) Overall order of the reaction
(d) Stoichiometry of the balanced equation

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Answer: (c) Overall order of the reaction — Units of k = (concentration)^(1−order) time^{-1}, so they depend on overall order.

XIII. A first-order reaction has half-life 20 minutes. The rate constant is:

(a) 0.05 min^{-1} **(b)** 0.693 min^{-1}
(c) 0.0347 min^{-1} **(d)** 13.86 min^{-1}

Answer: (c) 0.0347 min^{-1} — k = 0.693/t½ = 0.693/20 = 0.0347 min^{-1}.

Q.2 — Short Answer Questions

a. What do you understand by the rate of a reaction?

The rate of reaction is the change in concentration of a reactant or product per unit time. It is expressed as −Δ[reactant]/Δt or +Δ[product]/Δt, usually in mol dm^{-3} s^{-1}.

b. Give the difference between enthalpy change of reaction and energy of activation.

Enthalpy change (ΔH) is the net heat difference between products and reactants; it determines whether a reaction is exo/endothermic. Activation energy (Ea) is the minimum energy barrier reactants must overcome to react; it determines the rate. A catalyst lowers Ea but does not change ΔH.

c. Differentiate clearly between order and molecularity of a reaction.

Feature	Order	Molecularity
Definition	Sum of powers in the experimental rate law	Number of species colliding in an elementary step
Determined by	Experiment	Reaction mechanism (elementary step)
Values	Can be 0, fractional, negative	Always a positive integer (1, 2, 3)
Applies to	Overall or elementary reactions	Elementary steps only

d. Why does the instantaneous rate change during a reaction?

Instantaneous rate depends on reactant concentrations at that moment. As reactants are consumed, their concentrations decrease, so the instantaneous rate usually decreases with time.

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e. Briefly summarize the effects of temperature and surface area on reaction rates.

Increasing temperature increases the rate because more molecules have energy ≥ Ea and collisions are more frequent. Increasing surface area (e.g. powdering a solid) exposes more particles to collisions, increasing the rate of heterogeneous reactions.

f. Justify that radioactive decay is always a first order reaction.

The decay rate is directly proportional to the number of undecayed nuclei present: rate = kN. Each nucleus decays independently with a constant probability, so the process follows first-order kinetics and has a constant half-life.

g. A reaction is second order with respect to a reactant. How is the rate affected if concentration is doubled and reduced to half?

Rate ∝ $[A]^2$. Doubling [A] increases the rate $2^2 = 4$ times. Halving [A] reduces the rate to $(½)^2 = ¼$ of the original.

h. What is meant by half-life and what is it used for?

Half-life ($t_{½}$) is the time required for the concentration of a reactant to fall to half its initial value. It is used to characterise reaction speed, compare rates, and (for first-order processes) calculate the rate constant: $k = 0.693/t_{½}$.

i. Why does wood burn more rapidly in pure oxygen than in air?

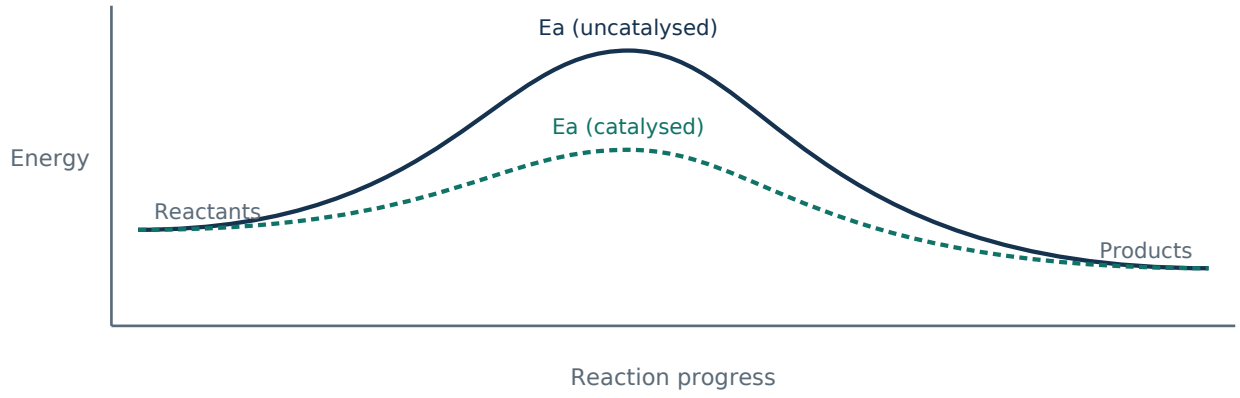
Pure oxygen has a much higher O_2 concentration than air (~21% O_2). More frequent effective collisions with oxygen increase the combustion rate.

j. A catalyst lowers the activation energy of a chemical reaction. Illustrate it.

A catalyst provides an alternative pathway with a lower energy barrier. On an energy profile, the catalysed curve has a lower peak than the uncatalysed curve, while reactant and product energy levels (ΔH) remain unchanged.

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Activation energy: uncatalysed vs catalysed



k. The rate constant for a reaction is 3.5×10^{-4} s^{-1} at 25 °C. What is the order? Explain from the units.

The unit s^{-1} (time^{-1} only) corresponds to a first-order reaction, because for first order k has units of time^{-1}.

l. If the initial concentration of the reactant is 0.50 mol dm^{-3}, calculate the initial rate of the reaction.

Taking the reaction in part (k) to be first order with k = 3.5×10^{-4} s^{-1}: rate = k[A] = $(3.5 \times 10^{-4})(0.50) = 1.75 \times 10^{-4}$ mol dm^3 s^{-1}.

Textbook check (p. 161): no rate law is restated here; the calculation assumes the first-order reaction and k from part (k).

m. How would the rate of this reaction change if the concentration of the reactant were doubled?

For the first-order reaction assumed above, rate ∝ [A], so doubling the concentration doubles the rate.

The question refers to “this reaction” without restating the rate law; the answer uses the first-order assumption from parts (k)–(l).

n. A first-order reaction has k = 2.5×10^{-3} s^{-1}. Calculate the half-life in minutes.

$t_{½} = 0.693/k = 0.693/(2.5 \times 10^{-3}) = 277.2$ s = 277.2/60 = 4.62 min.

o. A radioactive isotope decays by a first-order process with half-life 12 hours. Calculate the rate constant in s^{-1}.

$k = 0.693/t_{½} = 0.693/(12 \times 3600\ s) = 0.693/43200 = 1.60 \times 10^{-5}$ s^{-1}.

Descriptive / Long Questions

Q.3 Relate the order of a reaction to the rate law. How do you distinguish between zero, first and second order reactions?

For rate = k[A]^m[B]^n, the overall order is m + n. The order is the sum of the exponents in the experimental rate law.

Order	Rate law	Units of k	Effect of doubling [A]	$t_{½}$
Zero	rate = k	mol dm^{-3} s^{-1}	No change	$[A]_0/2k$ (depends on $[A]_0$)
First	rate = k[A]	s^{-1}	Doubles	0.693/k (constant)
Second	rate = $k[A]^2$	dm^3 mol^{-1} s^1	Quadruples	$1/(k[A]_0)$

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Q.4 How do you find the numerical value of a rate constant by initial-rate and half-life methods?

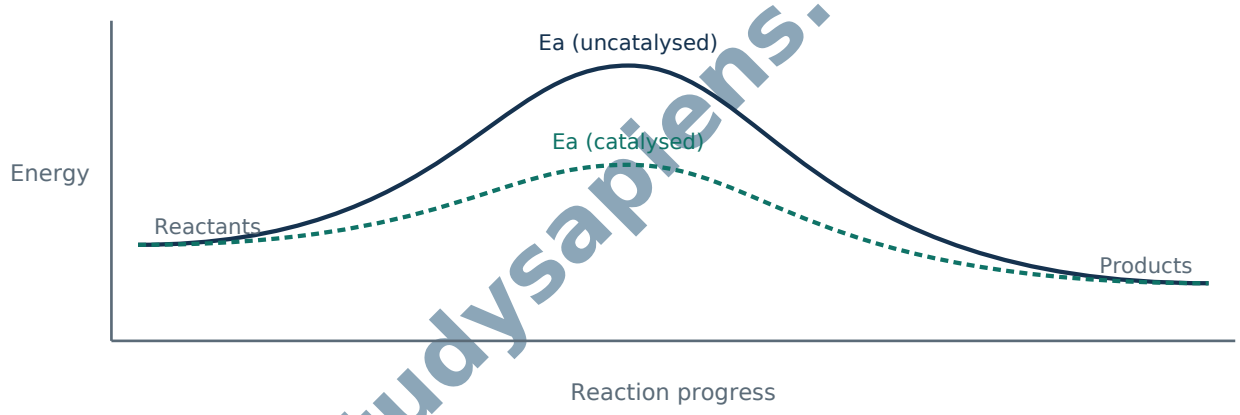
Initial-rate method: measure the initial rate for different initial concentrations. Compare experiments to find the order with respect to each reactant, then substitute one set of values into rate = k[A]^m[B]^n to calculate k.

Half-life method: for a first-order reaction, measure t½ and use k = 0.693/t½. For other orders, the dependence of t½ on initial concentration reveals the order (e.g. $t_{½} \propto 1/[A]_0$ for second order).

Q.5 How does the activation energy profile of an uncatalysed reaction compare with that of a catalysed reaction?

Both profiles start at the same reactant energy and end at the same product energy, so ΔH is unchanged. The uncatalysed reaction has a higher peak (larger Ea). The catalysed reaction follows an alternative pathway with a lower peak (smaller Ea), so a larger fraction of molecules can react and the rate increases.

Activation energy: uncatalysed vs catalysed



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Q.6 The reaction between H_2O_2 and I^- in acidic solution occurs via: Step 1 $H_2O_2 + I^- \rightarrow H_2O + OI^-$ (slow); Step 2 $OI^- + H^+ \rightarrow HOI$ (fast); Step 3 $HOI + I^- + H^+ \rightarrow H_2O + I_2$ (fast). (i) Overall equation. (ii) Intermediates and catalysts. (iii) Rate-determining step. (iv) Rate equation.

(i) Adding the steps and cancelling intermediates: $H_2O_2(aq) + 2H^+(aq) + 2I^-(aq) \rightarrow 2H_2O(l) + I_2(aq)$.

(ii) Intermediates: OI^- and HOI (formed then consumed). There is no catalyst; I^- is a reactant consumed overall.

(iii) The rate-determining step is Step 1 (the slow step).

(iv) Rate = $k[H_2O_2][I^-]$, because the slow step involves one H_2O_2 and one I^-.

Numerical Problems (fully worked)

Q.6 Calculate the reaction rate if [A] = 0.5 M, [B] = 0.2 M and k = 4.0 M^{-2} s^{-1}. Rate law: rate = k[A][B]².

Given: [A] = 0.5 M; [B] = 0.2 M; k = 4.0 M^{-2} s^{1}

Formula: rate = k[A][B]²

rate = 4.0 × 0.5 × $(0.2)^2$ = 4.0 × 0.5 × 0.04 = 0.080 mol dm^{-3} s^{-1}.

Answer: Rate = 0.080 mol dm^{-3} s^{1}

This is the second Q.6 in the book (Numerical Problems); the descriptive Q.6 is answered above.

Q.7 A first-order reaction has k = 5.5 × 10^{-14} s^{-1}. Find the half-life.

Given: k = 5.5 × 10^{-14} s^{-1}

Formula: $t_{½}$ = 0.693/k

$t_{½}$ = 0.693/(5.5 × 10^{-14}) = 1.26 × 10^{13} s (≈ 4.0 × 10^{5} years).

Answer: $t_{½}$ ≈ 1.26 × 10^{13} s

Q.8 For 2HI(g) → H_2(g) + I_2(g), three experiments give [HI] (M) and rate (M/s): 0.015 → 1.1×10^{-3}; 0.030 → 4.4×10^{-3}; 0.045 → 9.9×10^{-3}. Write the rate law and find k with units.

Given: [HI] = 0.015, 0.030, 0.045 M; rates = 1.1×10^{-3}, 4.4×10^{-3}, 9.9×10^{-3} M s^{-1}

Formula: rate = k[HI]ⁿ; compare experiments to find n

Doubling [HI] (0.015→0.030) quadruples the rate (1.1→4.4×10^{-3}), so n = 2. Rate = k[HI]².

k = rate/[HI]² = (1.1×10^{-3})/$(0.015)^2$ = 1.1×10^{-3}/2.25×10^{-4} = 4.9 M^{-1} s^{-1}.

Checking experiment 2: 4.4×10^{-3}/$(0.030)^2$ = 4.9 M^{-1} s^{1} (consistent).

Answer: Rate = k[HI]²; k ≈ 4.9 M^{-1} s^{1} (dm^3 mol^{-1} s^{-1})

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Quick Check Solutions (in-chapter practice boxes)

Quick Check 7.1 (book p. 141)

a. What role does activation energy play in chemical reactions?

It is the minimum energy barrier that colliding particles must overcome to form the activated complex and react.

b. How does activation energy affect the rate of a reaction?

A higher activation energy means fewer molecules have enough energy to react, so the rate is slower; a lower activation energy gives a faster rate.

Quick Check 7.2 (book p. 143)

For $H_2(g) + I_2(g) \rightarrow 2HI(g)$, $[I_2]$ falls from 0.010 to 0.0080 mol dm^{-3} in 100.0 s. Find the average rate.

Average rate = $-\Delta[I_2]/\Delta t = (0.010 - 0.0080)/100.0 = 2.0 \times 10^{-5}$ mol dm^3 s^{-1}.

Quick Check 7.3 (book p. 144)

a–c. Plot Table 7.1 data for HI and find rates by drawing tangents at 300 s (0.03 M) and at 0.10, 0.050, 0.02 mol dm^{-3}.

Plot [HI] against time and draw a smooth curve. The instantaneous rate at any point is the magnitude of the slope of the tangent at that point: rate = $-$d[HI]/dt. (Numerical values require the Table 7.1 data.)

d. What do you deduce about the rate with time?

The rate decreases with time as the concentration of HI falls.

e. At which concentration is the rate highest and lowest?

Highest at the highest concentration (0.10 mol dm^{-3}); lowest at the lowest concentration (0.02 mol dm^{-3}).

Quick Check 7.4 (book p. 147)

a. What is the Boltzmann distribution curve?

A graph showing the distribution of kinetic energies among molecules in a sample; the area under the curve equals the total number of molecules, and the area beyond Ea gives the fraction able to react.

b. Why a 10 °C rise approximately doubles the rate.

The fraction of molecules with energy ≥ Ea increases exponentially with temperature (Maxwell–Boltzmann distribution), so many more effective collisions occur; for many reactions this roughly doubles the rate.

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Quick Check 7.5 (book p. 149)

a. Can a catalyst be consumed in a chemical reaction? Why or why not?

No. A catalyst participates in intermediate steps but is regenerated at the end, so it is not consumed overall.

b. Is $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$ with $V_2O_5(s)$ heterogeneous or homogeneous catalysis?

Heterogeneous catalysis: the catalyst V_2O_5 is a solid while the reactants are gases (different phases).

c. Draw an energy profile for uncatalysed and enzyme-catalysed reactions, showing Ea for both.

Both curves start and end at the same energy levels (same ΔH). The uncatalysed curve has a higher peak (larger Ea); the enzyme-catalysed curve has a lower peak (smaller Ea).

Quick Check 7.6 (book p. 151)

a. How is the order of reaction derived from the rate law?

The overall order is the sum of the exponents of the concentration terms in the experimental rate law.

b. What is the specific rate (rate constant) and how is it represented?

The rate constant k is the proportionality constant in the rate equation (rate = k[A]^m[B]^n). It equals the rate when all reactant concentrations are unity and depends on temperature.

Quick Check 7.7 (book p. 154)

a(i). For rate = $k[NO]^2[NH_3]^0$ and rate = $k[BrO_3^-][Br^-][H^+]^2$, calculate the overall order.

First expression: 2 + 0 = 2 (second order). Second expression: 1 + 1 + 2 = 4 (fourth order).

a(ii). Orders with respect to each reactant.

First: NO = 2, NH_3 = 0. Second: BrO_3^- = 1, Br^- = 1, H^+ = 2.

c. Why the sum of coefficients of a balanced equation is not necessarily the order.

Order is determined experimentally from the rate law and reflects the mechanism (especially the slow step), not the overall stoichiometric coefficients.

Quick Check 7.8 (book p. 156)

A first-order reaction has $t_{½}$ = 15 min and initial concentration 0.100 mol dm^{-3}. Calculate k.

For first order, $k = 0.693/t_{½} = 0.693/15 = 0.0462\ min^{-1}$ (independent of initial concentration).

Quick Check 7.9 (book p. 158)

Explain why the proposed mechanism for $H_2O_2 + 2H^+ + 2I^- \rightarrow 2H_2O + I_2$ is consistent with rate = $k[H_2O_2][I^-]$.

The slow (rate-determining) step is $H_2O_2 + I^- \rightarrow H_2O + IO^-$. Its rate law is rate = $k[H_2O_2][I^-]$, which matches the observed rate equation. The subsequent fast steps do not affect the overall rate.

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Textbook checks flagged in this chapter

• p. 161: Q.2(l) and (m) do not restate a rate law; the solution assumes the first-order reaction and k from part (k).

• p. 161–162: the number Q.6 is used twice (Descriptive and Numerical); both are answered.

• p. 162: a k-versus-temperature table is printed without a question stem; no calculation is requested for it.

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